

AI-BFSI Internship Project Documentation

Data Flow Diagram – Subscription Waste Detector

1. Purpose

This document explains the logical movement of data within the **Subscription Waste Detector**. It identifies external entities, processing modules, data stores, and outputs. The goal is to provide a clear technical reference for engineers and a visual explanation for stakeholders on how expense information becomes actionable subscription insights.

2. Executive Summary

The Subscription Waste Detector converts raw expense records into structured financial insights. Users upload expense data via **CSV, Excel, or manual entry**. Python and Pandas validate and process the records, recurring subscriptions are detected, dashboards are generated, and the **Groq API** provides optimisation recommendations.

This sequential pipeline ensures that every stage adds value, transforming raw data into meaningful decision-support information.

3. Context-Level Diagram

External Entity: User **System:** Subscription Waste Detector

Inputs:

- CSV File
- Excel File
- Manual Entry

Outputs:

- Dashboard
- Subscription Summary
- AI Recommendations

4. Level 1 Data Flow

1. **Expense Upload** Users provide expense records via CSV, Excel, or manual entry.
2. **Data Validation** Uploaded data is checked for accuracy and consistency.
3. **Python & Pandas Processing** Records are organised and recurring transactions analysed.
4. **Subscription Detection** Identifies recurring payments and categorises them.

- 5. **Dashboard Generation** Insights are presented visually through interactive dashboards.
- 6. **Groq AI Suggestions** AI generates cost-saving and optimisation recommendations.

5. Data Components

- **External Entity:** User
- **Data Store:** Expense Dataset (in-memory/session-based)
- **Processing Modules:** Validation, Python Processing, Pandas Analysis, Subscription Detection, Dashboard Generation, AI Engine
- **Outputs:** Subscription Summary, Spending Charts, Renewal Reminders, Cost-Saving Suggestions

6. Data Dictionary

Data Element	Description	Used In
Expense Record	Individual financial transaction supplied by user	Data Processing
Category	Classification of transaction	Dashboard
Subscription	Recurring payment identified	Detection Module
Renewal Date	Upcoming payment date	Reminder Module
Monthly Spending	Aggregated subscription expenditure	Dashboard Charts
AI Suggestion	Cost optimisation recommendation (Groq API)	AI Module

7. Conclusion

The Subscription Waste Detector follows a **sequential data flow pipeline** where each stage transforms raw expense records into structured insights. This reduces manual analysis, improves subscription visibility, and empowers users to make better financial decisions.

Key Takeaways:

- Multiple input methods improve usability.
- Python & Pandas automate subscription analysis.
- Dashboards provide clear financial insights.
- Groq AI enhances decision-making with intelligent recommendations.